

The Effects of Big Data On the Business Model of McKee Foods Corporation and How Other

Companies Can Benefit

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MGT 5810 Information Technology For Business

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Abstract

Through the implementation and use of big data, web usage, computing, and software development at McKee Foods Corporation in Collegedale, Tennessee, this privately-traded and family-owned corporation has led the multiple brands of products at McKee Foods such as Drake's, Fieldstone Bakery, Heartland Brands, Little Debbie, and Sunbelt Bakery to develop products that are rare, valuable, hard to imitate, and hard to replace in the baked sweet goods market of the United States and particularly the southeastern United States such as Tennessee. After a thorough analysis, it can be concluded that these technologies of big data, web usage, computing, and software development have positive side effects and negative side effects to the business model of McKee Foods, but ultimately implementing these technologies into the business model of the corporation and utilizing these technologies in the right manner can lead the corporation to successfully take advantage of the best of what these technologies have to offer and achieve the company statement or mission of the organization. Applying these technologies in a similar manner the ways McKee Foods has used big data, web usage, computing, and software development as well as recognizing the benefits and costs of implementing this technology within a business model of a relatively large corporation or organization can help other corporations reap the benefits of what these technologies have to offer.

Keywords: McKee Foods Corporation, Big Data, Web Usage, Computing, Software Development, Information Systems, Management Information Systems, Information Technology

The Effects of Big Data On the Business Model of McKee Foods Corporation and How Other Companies Can Benefit

When a corporation decides to implement big data, web usage, computing, and software development into its business model, a thorough analysis must be conducted to analyze the pros and cons of implementing this software into the business model so the business model achieves the ultimate mission or vision statement of the organization. The corporation McKee Foods headquartered in Collegedale, Tennessee and seller of the brands Drake's, Fieldstone Bakery, Heartland Brands, Little Debbie, and Sunbelt Bakery in the baked sweet goods market of the United States can be analyzed as an example of a privately-traded and family-owned corporation that has leveraged big data, web usage, and software development to expose the pros and cons of what these technologies have to offer, and how a large corporation in the United States can utilize these technologies within their business model to make intelligent decisions to achieve the set explicit goals of the corporation. Using technologies that utilize big data from point-of-sale data in grocery stores, convenient stores, and other organizations that sell McKee Foods brands product in the United States through programs and software such as Nielsen and JDA space management planning software can help organizations such as McKee Foods reach decisions based on customer demographics and buying behaviors that might expose the strengths and weaknesses of the corporations marketing plan. Analyzing how the impacts of these current technologies in the present can potentially help many corporations in the United States proactively recognize what might happen in the future, and effectively prepare for the benefits the technology might bring as well as prepare for otherwise unexpected costs to the company that might negatively impact the organizations set mission or goal.

Description of the Organization

McKee Foods Corporation is headquartered in Collegedale, Tennessee and has operations located in Collegedale, Tennessee, Gentry, Arkansas, Stuarts Draft, Virginia, and Kingman, Arizona (McKee Foods, n.d.). The company has 6,300 employees who work at the three plants in Collegedale, Tennessee, Gentry, Arkansas, and Stuarts Draft, Virginia and a distribution center in Kingman, Arizona. The employees make up mostly plant employees at the Collegedale, Gentry, and Stuarts Draft locations but also make up office employees who work in finance, marketing, human resources, food science, engineering, and logistics at the Collegedale and Kingman locations. The role of information technology is used in all areas of the corporation and plays a major role in the operations of McKee Foods through the communications, marketing, accounting, finance, and distribution processes of the corporation. The corporation uses information technology 24 hours a day, 7 days a week as most of the plants rarely stop operations throughout the year, and information technology creates an efficient process for all 5 brands of baked sweet goods to be marketed, produced, and distributed in a timely and effective manner to retailers and convenient stores across the United States.

Discussion

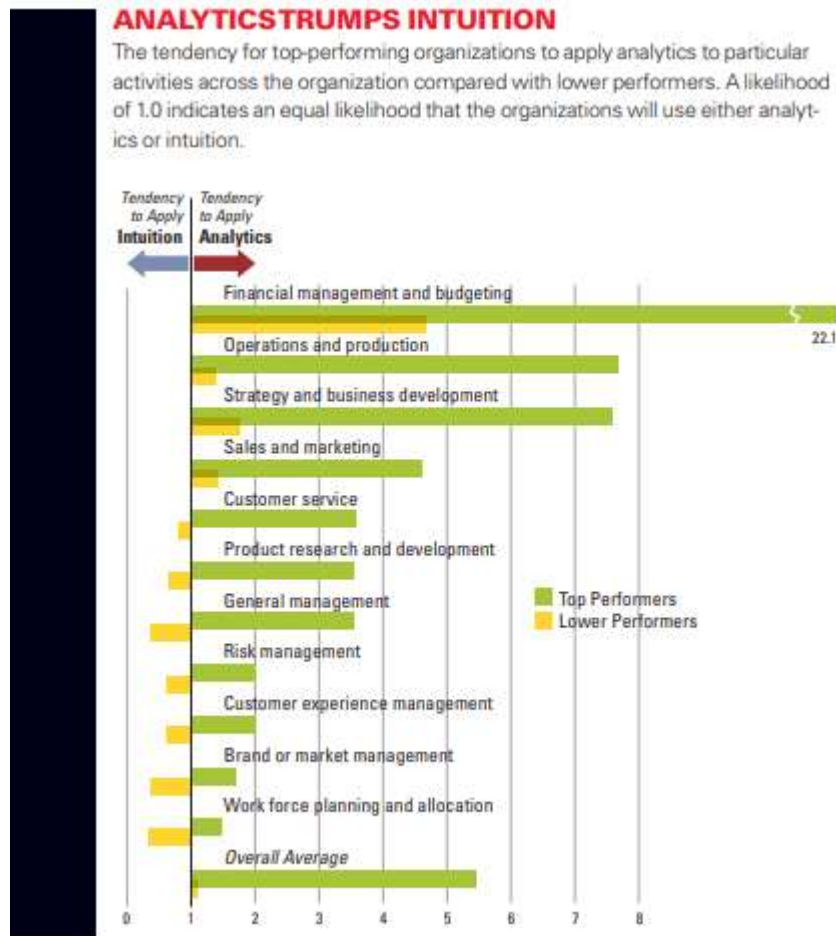
With the rise of many large corporations in the United States using more technology-based tools in their business models, the application of big data, web usage, computing, and software development is crucial for large corporations such as McKee Foods located in Collegedale, Tennessee to maintain competition in the baked sweet goods market of the United States, and it is also important for other corporations to recognize the importance of the implementation of these technologies as markets develop and competition grows between tech-based companies that also rely heavily on these technologies. The level of competition and competitiveness to implement the most effective information technology in the business model of a company could have first been clearly illustrated through the “war” of Microsoft vs. Apple and Bill Gates vs. Steve Jobs from 1978-2007 when many considered Steve Jobs the ultimate winner by outperforming Microsoft and developing the iPhone in 2007 (Bill Gates vs. Steve Jobs, n.d.). Technology columnist Kara Swisher commented on the outperformance of Apple over Microsoft and how Apple dominated the market through mastering information technology:

The concept that you carry around and hold these supercomputers in your hand, and do all kinds of things with them and interact with the world is a big idea. You had to pick one person who impacted technology more than anyone in this era, it would have to be Steve Jobs. (Bill Gates vs. Steve Jobs, n.d.)

The importance and necessity of information technology to survive in a tech-dominated market was also illustrated with the competition of Netflix vs. Blockbuster when Netflix demonstrated it had assets that are valuable, rare, imperfectly imitable, and non-substitutable by forcing Blockbuster into bankruptcy (Gallaughier, 2018). Although Blockbuster adapted to the relatively

rapid-changing market of information technology and tried to imitate the Netflix DVD subscription service, the company could not maintain a sustainable competitive advantage like Netflix and could not sustain a product that is rare, valuable, hard to imitate, and hard to replace. Now the large corporation Netflix that employed thousands of people is only a reminder of how powerful information technology can be. Although McKee Foods Corporation is a smaller, private and family-owned corporation in Collegedale, Tennessee, the same principles apply and taking advantage of big data and software development is still important for the success of the company because these examples illustrate how quickly a large corporation with thousands of employees can lose market share, go bankrupt, and even disappear from the market because they weren't able to sustain a product that is rare, valuable, hard to imitate, and hard to replace.

According to a review of the literature on the impact and importance of the implementation of big data and information technology, big data is one of the most important factors in decision making, and there seems to be a connection between top-performing organizations that apply analytics to their decision making rather than relying on their own intuition to make important decisions within the organization. According to an article titled *Big Data, Analytics, and the Path From Insights to Value* from the MIT Sloan Management Review in 2011, top performers of organizations had a significantly higher likelihood of relying on big data to make decisions rather than by relying on intuition (LaValle, Lesser, Shockley, Hopkins, & Kruschwitz, 2010):



The study also found organizations that were top-performing were twice as likely to use big data analytics to guide day-to-day operations in comparison to lower performing organizations who do not (LaValle et al., 2010). This likely could be because these organizations are able to create a product that is rare, valuable, hard to imitate, and hard to replace using big data and are able to create a sustainable competitive advantage over other organizations who do not use big data to influence decision making and as a result, these other organizations are not able to create a product that is rare, valuable, hard to imitate, and hard to replace.

Using big data tools such as Nielsen and JDA space planogram software has given McKee Foods the ability to recognize target markets of their brands and also further develop

their brands and products to create a product that is rare, valuable, hard to imitate, and hard to replace through implementing big data in decision making. I experienced the effects of using Nielsen and JDA to make decisions as an intern on the category insights team at McKee Foods where I personally worked with the category insights team to look at big data pulled from Nielsen to draw conclusions based on sales in geographic regions on products such as the Little Debbie Oatmeal Cream Pie. On one of my first days of training on how to use Nielsen at McKee Foods, I was asked to rely on intuition and make guesses where the Little Debbie Oatmeal Cream Pie had higher sales in particular states across the United States. Based on my intuition and knowledge of McKee Foods and its involvement in nearby communities, I made the correct guess the Little Debbie Oatmeal Cream Pie sells more in Tennessee in comparison to other states in the United States, but by looking deeper into the information from Nielsen I realized I might not be able to rely solely on my intuition to make decisions. One surprising conclusion I gathered from the Nielsen marketing data but was unable to gather from my own intuition was the Little Debbie Oatmeal Cream Pie product sold relatively well in southern states such as Tennessee but did not sell as well in northern states, such as Minnesota and New York, and was competitive with the product Entemmanns Little Bites there.

Using Nielsen also gave the category insights team and myself the ability to recognize the demographics of customers, and the ability to continually develop the Little Debbie Oatmeal Cream Pie into a product that is rare, valuable, hard to imitate, and hard to replace. Looking at the data pulled from Nielsen, I could make conclusions on which demographics of customers enjoyed which brands and products and in what regions of the United States. For example, I came to the conclusion after looking at overall sales of the Little Debbie Oatmeal Cream Pie that this product was particularly popular among the Hispanic population, and was also popular at

HEB grocery stores in South Texas. Again, this was an insight I could not reach from my own intuition but was able to reach looking deeper into the data, and I was able to make conclusions from concrete finite numbers on the sales of the Little Debbie Oatmeal Cream Pie among Hispanics at HEB stores in South Texas.

Using this data on what demographics of customers bought the Oatmeal Cream Pie and in what locations in the United States the Oatmeal Cream Pie seemed to sell relatively well in gave our category insights team the ability to further develop products that might be popular with Hispanic customers and also gave us the ability to market products in geographic regions where we could improve sales, making the Little Debbie Oatmeal Cream Pie a product that is rare, valuable, hard to imitate, and hard to replace. Looking at the data pulled from Nielsen gave us the conclusions the Little Debbie Oatmeal Cream Pie had a relatively high amount of what Professor John Gallagher calls staying power (Gallagher, 2018). We came to these conclusions because we felt the data pulled from Nielsen showed there were many customers in Tennessee who were loyal to the Little Debbie Oatmeal Cream Pie product, and many customers in Tennessee likely saw it as a product of nostalgia because the Little Debbie brand has had a relatively high amount of market share and prominence in Tennessee since the creation of McKee Foods in Chattanooga, Tennessee in 1934 (McKee Foods, n.d.). Using these conclusions drawn from Nielsen shopper marketing data, this indicated to our team we may need to focus more on a market development strategy of the product market growth matrix to areas where the Little Debbie Oatmeal Cream Pie was not as popular, such as New York and Minnesota (Gallagher, 2018). Using this strategy could also help us develop our Little Debbie brand to appeal more to customers in the northern United States which would likely make the Little Debbie brand rare, valuable, and hard to imitate in states such as New York and Minnesota. An

example of how McKee Foods used this strategy is the development of the Little Debbie Little Muffins product, which seems to be more popular with customers in northern states such as Minnesota and New York. Looking at the shopper marketing data from Nielsen revealed to us that one of the best-selling products in the baked sweet goods market in northern states such as Minnesota and New York is a product made by one of McKee Foods biggest competitors, Entemann's Little Bites Muffins. Developing a product similar to Entemann's Little Bites to market and sell to customers in New York and Minnesota led McKee to develop Little Debbie Little Muffins. The product seems rare, valuable, hard to imitate and hard to replace enough so that Little Debbie Little Muffins are able to gain a significant amount of market share in northern states. If McKee Foods is able to maintain the quality of the Little Debbie Oatmeal Cream Pie product in southern states such as Tennessee, I think the staying power of this product will only grow as the nostalgia of this product grows with customers and the product will remain one that is rare, valuable, hard to imitate, and hard to replace. This will contribute to McKee's sustainable competitive advantage over competitors in the baked sweet goods industry such as Entemanns in Tennessee.

Like Nielsen shopper marketing data, JDA space planning software also used big data and was able to create and sustain products at McKee such as the Little Debbie Oatmeal Cream Pie that is rare, valuable, hard to imitate, and hard to replace. As an intern on the category insights team at McKee Foods, I used the JDA space planning software program on my desktop computer and laptop to organize the shelf space and different brands of McKee Foods and competitor brands of McKee such as Entemanns, Tastykake, Bimbo, and Sara Lee for one of McKee Foods largest customers, Wal-Mart. Using JDA space planning software, I was able to look at analytics such as days of sale outstanding and overall sales of McKee Foods products as

well as products of competitors to make decisions on where the products should be placed on shelves of hundreds of Wal-Marts across the United States using a non-biased opinion.

Analyzing this data gave me the ability to make judgements and place products that sell quickly such as the Little Debbie Oatmeal Cream Pie and Entemann's Little Bites on a shelf that is close to customers eye-level and products that do not sell as quickly such as Sunbelt Bakery Granola Bars on shelves near the bottom where customers are less likely to notice and buy them. Using this big data in the JDA space management software program to organize the shelf space for one of the largest retailers in the United States likely gave McKee Foods the ability to make McKee Foods brand products rare, valuable, hard to imitate, and hard to replace because this has likely given McKee Foods an established relationship with Wal-Mart that is rare and valuable. This likely has also made McKee Foods brands more difficult to imitate and harder to replace because of that relationship they have built with Walmart. It would likely be difficult for another corporation or organization to try to imitate McKee Foods Little Debbie Oatmeal Cream Pie and establish the reputation McKee Foods has built with Wal-Mart to have the opportunity to place its product on the shelf in the baked sweet goods section of almost every Wal-Mart in the United States. This possibly was an unforeseen benefit of investing in data analytics by McKee and employees who were knowledgeable of big data for McKee Foods, but as a result big data has given McKee Foods the opportunity to establish and maintain a strong bond with one of the biggest retailers in the United States, and maintain a relationship with possibly one of the most desirable retailers to have a relationship with in the baked sweet goods market of the United States.

Utilizing software development to further the use of big data in the JDA space planning software program would have likely made the process even more effective and less time

consuming while I was on the category insights team at McKee Foods. Because our manpower on the category insights team was limited to about a dozen employees at a time, and because Wal-Mart held the category insights team responsible for organizing the shelf space for nearly every individual Wal-Mart in the United States, the project was time consuming and took several months to complete and had to be completed again every year due to sales changing every year. I noticed it also seemed to dampen employee morale because of the tedious task of having to use JDA and individually sort and pick which products would be most appropriate for which shelf. Not only was the task tedious and time consuming for the entire category insights team, but sorting and picking products to different shelves were judgements made by different employees of the category insights team and there seemed to be no “right” or “wrong” answers. Consequently, many JDA planograms were rejected by Wal-Mart because Wal-Mart did not always agree with the judgements of our team members on the category insights team on which products belonged to which. I remember wishing our team had software engineers to automate the process, and to possibly write a code or algorithm that could automatically organize the shelf space based on the data analytics we were trying to analyze and make decisions from. After reading *The 4 Hour Work Week* by Tim Ferriss these past couple of weeks, I have been thinking of more ways to automate my life and have been thinking of how I could have used critical thinking to automate the processes I have done at work in the past, and the main process that comes to mind is when I was on the category insights team at McKee and was spending most of my days in the JDA Space Planning software program trying to organize shelf space for Wal-Mart. Using software development to improve the big data we were using in the JDA Space Planning program could have saved our category insights team hundreds if not thousands of hours spent, and could have saved McKee Foods thousands of dollars. These resources used for

the Walmart project could have been spent towards big data that would have contributed to making the brands of McKee Foods rare, valuable, hard to imitate, and hard to replace.

For a corporation to successfully implement big data and software development into their business model using tools such as Nielsen and programs such as JDA Space Planning, a corporation must recognize the potential costs of implementing big data and software development before implementing these technologies, such as the side effect of possibly having a less social and collaborative work force within the business model of the organization or corporation. According to a study published in the Harvard Business Review, a Gallup sample showed that the use of social media was negatively associated with certain measurements of participants well-being, such as life satisfaction, self-reported mental health, self-reported physical health, and body-mass index (Christakis, 2017). Even many notable public figures such as Warren Buffet have avoided the iPhone revolution and use flip phones that were developed before the release of the first iPhone in 2007 (Malito, 2018). According to an article titled *Positive Social Interactions and the Human Body at Work: Linking Organizations and Physiology* in the Academy of Management Review, human physiological systems are highly responsive to positive social interactions and there seems to be many positive physiological benefits of human interaction between employees within an organization that might be prevented by the development and increased use and reliance on information technology (Heaphy & Dutton, 2006). After recently reading the book *Tribe* by New York Times bestselling author Sebastian Junger, I have been convinced human beings tend to react positively in response to human interaction between others and seem to work better with others when having to problem solve and make decisions on difficult problems within a group or team. In the book, Junger uses first-hand experiences as a journalist in Afghanistan following an American Infantry platoon in

some of the most hostile areas in Afghanistan during the conflict between the United States and Afghanistan in 2007 to make the argument that human beings tend to desire interaction with other human beings in times of conflict and hardship when critical decisions need to be made within an organization or group. Collaborating with Wal-Mart to make progress for the JDA space planning project at McKee Foods with the category insights team gave me the insight that one of the major side effects of using mostly technology such as video conferencing tools to communicate with Wal-Mart was there seemed to be little face-to-face interaction between the category insights team and the team we were communicating with at Wal-Mart. As a result, this seemed to make the task of managing big data and completing the project more difficult. Not only were there issues in communicating deadlines and communicating our personal emotions with the team at Wal-Mart, but the team at Wal-Mart seemed to show less sympathy and may not have identified with our team and bonded with our team as much than if we had more face-to-face communication. For example, if some of our employees were sick and did not come into the office on a particular day when our team was supposed to work on the JDA space planning project for Wal-Mart, the team at Wal-Mart may not have been as sympathetic to our situation, because it can likely be difficult for teams to have sympathy and feel an emotional connection to other teams that work in an office hundreds of miles away with little face-to-face communication, and it might have been difficult for Wal-Mart's team to understand why a deadline may be needed on a project. According to an article in the journal of politeness research, computer-mediated communication was found to provide varying degrees of anonymity that "may encourage a sense of impunity and freedom from being held accountable" (Hardaker, 2010). If more face-to-face interaction were used between employees of an organization and employees of other organizations, than these organizations may be able to lessen the negative side effects of less

face-to-face human interaction between employees of an organization and employees of an outside organization or corporation.

Another negative side effect that may arise from implementing big data within an organization's business model such as using shopper marketing data through Nielsen at McKee Foods is that using big data is complex and may require a significant amount of man power and knowledge to implement in the business model. I noticed the complexity of Nielsen on one of my first days as an intern at McKee Foods while I was being trained to use the data to draw conclusions and make decisions and while being tested on my knowledge. Not only did this task take a relatively highly experienced employee in Nielsen a full day to train me, but I was also continually training in the program and consulting with other full-time experienced employees on how to use and analyze Nielsen data for at least six months. Although I was experienced in Microsoft Excel and in similar programs that could analyze data, Nielsen usually requires one-on-one training and a significant amount of time and resources to master and use effectively. According to the article *Big Data, Analytics and the Path From Insights to Value* from the MIT Sloan Management Review, "lack of understanding of how to use analytics to improve the business" is the most common reason respondents answered as an obstacle to the widespread adoption of analytics in their organization with "lack of skills internally in the line of business" as the third most common reason (LaValle et al., 2010).

Finally, the last and potentially largest unforeseen negative side effect of implementing information technology such as big data, web usage, computing, and software development into the business model of an organization or corporation is the growing issue of tech-related jobs in areas such as big data and software development being mostly comprised of men. According to an online article from Forbes, women roughly only hold 24% of computer science jobs, in

comparison to 37% in 1995, and hold only 11% of executive positions in Silicon Valley (Howard, 2017). According to another article from ComputerWorld.com, 15,129 female students graduated with a bachelor's degree in computer science in 1986, but that number has continually declined since 1986 to 6,749 in 1996 (Mitchell, 2013). More recently in July of 2017 the issue of the amount of women in tech-related fields was brought to attention by corporations in the United States by Google engineer James Damore, who sent a memo to Google that discussed the issue of women in tech and hinted that one of the major factors there weren't as many women in tech as men is because of biological differences between men and women, and certain psychological differences between men and women such as extraversion and neuroticism as well as a higher variance of IQ among men. Other well-known figures in the United States have blamed the issue on sexism, unfavorable work environments, and social influences and biases in the work place that have discouraged women from entering into tech-related fields. It seems to be a trend across the United States for many companies that utilize information technology such as big data and software development to hire a disproportionate amount of men in comparison to women for tech-related jobs, and this could be a potential negative side effect of investing in information technology and hiring more employees for an organization or corporation that cares about workplace diversity. Recognizing the reasons why many women don't enter tech-related fields and finding solutions to encourage more women to enter the fields of information technology in areas such as big data, web usage, computing, and software development could possibly help an organization or corporation that heavily invests in information technologies avoid this possible side effect. Recruiting women into tech-related fields at local Universities and creating organizations specifically designed to encourage and recruit women into information technology fields may be a couple of many possible solutions. Creating organizations that also

educate women on the complex nature of information technology such as big data, web usage, computing, and software development could potentially help women become more educated in the field, so this side effect can be avoided by organizations and corporations who utilize these forms of technology. Maintaining awareness of this side effect along with the other side effects of using these technologies within the business model of a corporation will likely help a corporation experience the best of what information technologies have to offer.

Conclusions and Managerial Recommendations

Recognizing the benefits and potential costs of using big data, web usage, computing, and software development within a corporation such as McKee Foods in Collegedale, Tennessee is important for the organization to produce a product that is rare, valuable, hard to imitate, and hard to replace and to effectively use these information technologies so the best of what these technologies have to offer can be taken advantage of. Although analytics proved to be a reliable measure to make intelligent decisions at McKee Foods, there were many times at McKee Foods where the best decisions seemed to be made by analyzing the data and also relying on intuition to make a conclusion from data. I would recommend not solely relying on intuition, because although I could correctly guess the Little Debbie Oatmeal Cream Pie sold relatively well in southern states such as Tennessee, looking deeper into the data revealed to me that the Little Debbie Oatmeal Cream Pie did not sell as well in northern states such as New York and Minnesota. Entemanns Little Bites seemed to have a significant portion of the market share in these northern states. I would also recommend recognizing and preparing for the potential negative impacts of information technology to a corporation or organization before implementing these technologies, such as the time and resources required to use these technologies, as well as the possible side effects of less interpersonal communication between employees and recognizing there are less women than men that work tech-related jobs, and this may be an issue for some corporations or organizations. Recognizing these advantages as well as disadvantages will likely help corporations and organizations develop and maintain products that are rare, valuable, hard to imitate, and hard to replace and be successful in markets that are becoming more reliant on the power of information technology.

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